

RMO– 2014(MUMBAI)

Max Mark – 102

Time – 3 Hrs

Instructions:

- Calculators (in any form) and protractors are not allowed.
 - Ruler and compass allowed.
 - Answer all the questions.
 - All Questions carry equal marks.
 - Answer to each questions starts on a new page. Clearly indicate the question number.
1. Three positive real numbers a, b, c are such that $a^2 + 5b^2 + 4c^2 - 4ab - 4bc = 0$. Can a, b, c be the lengths of the sides of a triangle? Justify your answer.
 2. The roots of the equation $x^3 - 3ax^2 + bx + 18c = 0$ form a non-constant arithmetic progression and the roots of the equation $x^3 + bx^2 + x - c^3 = 0$ form a non-constant geometric progression. Given that a, b, c are real numbers, find all positive values a, b .
 3. Let ABC be an acute angled triangle in which $\angle ABC$ is the largest angle. Let O be its circumcentre. The perpendicular bisectors of BC and AB meet AC at X and Y respectively. The internal bisector of $\angle AXB$ and $\angle BYC$ meet AB and BC at D and E respectively. Prove that BO is perpendicular to AC if DE is parallel to AC.
 4. A person moves in $x - y$ plane moving along points with integer co-ordinates x and y only. When she is at point (x, y) , she takes a step based on the following rules:
 - (i) If $x + y$ is even she moves to either $(x + 1, y)$ or $(x + 1, y + 1)$;
 - (ii) If $x + y$ is odd she moves to either $(x, y + 1)$ or $(x + 1, y + 1)$
 How many distinct paths can she take to go from $(0,0)$ to $(8,8)$, given that she exactly three steps to the right (x, y) to $(x + 1, y)$?
 5. Let a, b, c be positive numbers such that $\frac{1}{1+a} + \frac{1}{1+b} + \frac{1}{1+c} \leq 1$. Prove that $(1+a^2)(1+b^2)(1+c^2) \geq 125$. When does the equality hold?
 6. Let D, E, F be the points of contact of the incircle of an acute angled triangle ABC with BC, CA, AB respectively. Let I_1, I_2, I_3 be the incentres of the triangle AFE, BDF, CED, respectively. Prove that the lines I_1D, I_2E, I_3F are concurrent.

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